CHEN, LEI

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Summary of achievements

- Published 25 papers in peer-reviewed journals and conferences (Google Scholar statistics: Citations (980), h-index (10), i10-index (12)), including *Nature Methods, Nature Communications*, *IEEE Transactions on Knowledge and Data Engineering, Briefing in Bioinformatics*, *Bioinformatics, NeurIPS, AAAI, SIGIR, EMNLP*, etc.
- Defined the current global performance standard of nanopore long-read small variant calling in competition with Google Health. My solutions are published in *Nature Methods, Nature Communications*, *Briefing in Bioinformatics,* and *Bioinformatics*, and are being used routinely in major bioinformatics centers and medical centers internationally, and locally.
- Awards and recognitions:
  - 2022, Shenzhen Universiade International Scholarship
  - 2021, Tencent Rhino-Bird Research Elite Training Program
  - 2022, Breakthrough Progress Award, Tencent
  - 2021, 2022, Dean Scholarship-Excellent & Special Award, SIAT
  - 2019, 2022, Outstanding Graduates Thesis & Graduates
  - 2018, National Scholarship for Encouragement
  - 2016, National Scholarship
- High-impact open-source software:
  - ClairS, ClairS-TO, Clair-Mosaic, Clair3-RNA, Gungnir

Research interests

1. AI for Science
2. Bioinformatics
3. Natural Language Processing
4. Recommender systems

Education

Ph.D. in Computer Science Sept 2022 – Aug 2026

The University of Hong Kong, Hong Kong
Supervisor: Prof. LUO, Ruibang

M.Eng. in Computer Technology Sept 2019 – Jun 2022

University of Chinese Academy of Sciences, Beijing
Affiliation: Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences
Supervisor: Prof. YANG, Min

B.Eng. in Computer Science and Technology Sept 2015 – Jun 2019

China University of Mining and Technology, Xuzhou

Experience

WeiXin Group (WXG), Tencent Jun 2021 – Jun 2022

2021 Tencent Rhino-Bird Research Elite Training Program, Research intern
Advisor: Dr. DING, Jingtao

Westlake University Apr 2021 – Aug 2021

Research intern
Advisor: Prof. YUAN, Fajie

Shenzhen Institute of Advanced Technology, CAS Sept 2018 – Aug 2019

Research intern
Advisor: Prof. YANG, Min

Awards and honors

1. 2022, Shenzhen Universiade International Scholarship
2. 2022, Breakthrough Progress Award, Tencent
3. 2021, 2022, Dean Scholarship-Excellent & Special Award, SIAT
4. 2021, 2022, Merit Student, UCAS
5. 2019, 2022, Outstanding Graduates Thesis & Graduates
6. 2016, 2017, 2018, The Zeng Xianzi Scholarship, CUMT
7. 2018, National Scholarship for Encouragement
8. 2017, The Sun Yueqi Excellent Student Award, CUMT
9. 2017, The First Prize Scholarship, CUMT
10. 2016, National Scholarship

Peer-reviewed publications (as of May 2026)

In my following listing, my publications are grouped into two categories:

- First-authored (^) and Co-first authored (#) publications
- Co-authored publications

Impact factors are as of May 2026. The full list of my publications is arranged and easily accessible from my Google Scholar profile at
https://scholar.google.com/citations?user=H25qJSMAAAAJ

First-authored (^) and Co-first authored (#) publications:

1. **L. Chen**^, Z. Zheng, J. Su, X. Yu, A. Wong, J. Zhang, Y. Lee, R. Luo. ClairS-TO: a deep-learning method for long-read tumor-only somatic small variant calling. ***Nature Communications (IF: 15.70)***, 16(1), 9630, 2026.
2. Z. Zheng, **L. Chen**#, J. Su, X. Yu, M. He, Y. Lee, T. Lam, R. Luo. ClairS: a deep-learning method for long-read tumor-normal pair somatic small variant calling. ***Nature Methods (IF: 32.10)***, 2026.
3. J. Zhang, **L. Chen**#, J. Sun, S. Li, Y. Zhou, ..., C. Li, Z. Zheng, R. Luo. Gungnir codec enabling high error-tolerance and low-redundancy DNA storage through substantial computing power. ***Nature Communications (IF: 15.70)***, 2026.
4. **L. Chen**^, J. Su, Z. Zheng, T. Lam, R. Luo. Large-scale Dataset and Effective Model for Variant-Disease Associations Extraction. ***ACM BCB***, 10.1145/3584371.3612995, 2023.
5. **L. Chen**^, J. Su, T. Lam, R. Luo. Exploring Pair-Aware Triangular Attention for Biomedical Relation Extraction. ***ACM BCB***, 10.1145/3584371.3612994, 2023.
6. **L. Chen**^, J. Ding, M. Yang, C. Li, C. Song, L. Yi. Item-provider co-learning for sequential recommendation. ***SIGIR (CCF A)*** (pp. 1817-1822), 2022.
7. **L. Chen**^, F. Yuan, J. Yang, X. He, C. Li, M. Yang. User-specific adaptive fine-tuning for cross-domain recommendations. ***IEEE Transactions on Knowledge and Data Engineering (IF: 10.40)***, 35(3), 3239-3252, 2021.
8. **L. Chen**^, F. Yuan, J. Yang, C. Li, M. Yang. Scene-adaptive Knowledge Distillation for Sequential Recommendation via Differentiable Architecture Search. ***WANT@ NeurIPS 2023***.
9. **L. Chen**^, F. Yuan, J. Yang, X. Ao, C. Li, M. Yang. A user-adaptive layer selection framework for very deep sequential recommender models. ***AAAI (CCF A)*** (Vol. 35, No. 5, pp. 3984-3991), 2021.

10. **L. Chen**^, R. Xu, M. Yang. Overview of the NLPCC 2020 shared task: multi-aspect-based multi-sentiment analysis (MAMS). ***NLPCC (CCF C)*** (pp. 579-585). Cham: Springer International Publishing. 2020.
11. **L. Chen**^, Z. Zheng, M. He, A. Wong, X. Yu, J. Li, J. Zhang, Y. Zhou, R. Luo. Clair-Mosaic: A deep-learning method for long-read mosaic small variant calling. bioRxiv, 2025-10. (Submitted to ***Genome Research***)

Co-authored publications:

12. Z. Zheng, X. Yu, **L. Chen**, Y. Lee, C. Xin, ..., R. Kesharwani, F. Sedlazeck, R. Luo. Clair3-RNA: A deep learning-based small variant caller for long-read RNA sequencing data. ***Nature Communications (IF: 15.70)***, 16(1), 11553, 2026.
13. Z. Zheng, M. He, X. Yu, J. Li, **L. Chen**, ..., J. Zhang, Y. Zhou, R. Luo. Accelerated long-read variant calling with Clair3 for whole-genome sequencing. ***Bioinformatics (IF: 5.40)***, btag181, 2026.
14. Z. Zheng, Y. Ren, **L. Chen**, A. Wong, S. Li, X. Yu, T. Lam, R. Luo. Repun: an accurate small variant representation unification method for multiple sequencing platforms. ***Briefings in Bioinformatics (IF: 7.70)***, 26(1), bbae613, 2025.
15. J. Zhang, Y. Zhou, Y. Ren, M. Au, K. Chow, **L. Chen**, Y. Zhao, J. Su, R. Luo. Toward owner governance in genomic data privacy with Governome. ***Cell reports methods (IF: 4.50)***, 5(9), 2025.
16. S. Li, F. Chow, **L. Chen**, J. Su, S. Ahmed, R. Luo. MirtronStructDB: A Comprehensive Database of Mirtrons with Predicted Secondary Structures. ***F1000Research***, 13, 899, 2024.
17. X. Zhang, **L. Chen**. MulMarker: a comprehensive framework for identifying multi-gene prognostic signatures. ***BIBM (CCF B)*** (pp. 4014-4021). IEEE. 2023.
18. G. Yuan, F. Yuan, Y. Li, B. Kong, S. Li, **L. Chen**, ..., Z. Li, Y. Xu, X. Qie. Tenrec: A large-scale multipurpose benchmark dataset for recommender systems. ***NeurIPS (CCF A)***, 35, 11480-11493, 2022.
19. Q. Jiang, **L. Chen**, W. Zhao, M. Yang. Toward aspect-level sentiment modification without parallel data. ***IEEE Intelligent Systems (IF: 6.10)***, 36(1), 75-81, 2021.
20. M. Yang, **L. Chen**, Z. Lyu, J. Liu, Y. Shen, Q. Wu. Hierarchical fusion of common sense knowledge and classifier decisions for answer selection in community question answering. ***Neural Networks (IF: 6.30)***, 132, 53-65, 2020.
21. Q. Jiang, **L. Chen**, R. Xu, X. Ao, M. Yang. A challenge dataset and effective models for aspect-based sentiment analysis. ***EMNLP (CCF B)*** (pp. 6280-6285), 2019.

22. M. Yang, W. Zhou, **L. Chen**, Q. Qu, Z. Zhao, Y. Shen. Investigating the transferring capability of capsule networks for text classification. ***Neural Networks (IF: 6.30)***, 118, 247-261, 2019.
23. M. Yang, **L. Chen**, X. Chen, Q. Wu, W. Zhou, Y. Shen. Knowledge-enhanced Hierarchical Attention for Community Question Answering with Multi-task and Adaptive Learning. ***IJCAI (CCF B)*** (pp. 5349-5355), 2019.
24. M. Yang, J. Liu, **L. Chen**, Z. Zhao, X. Chen, Y. Shen. An advanced deep generative framework for temporal link prediction in dynamic networks. ***IEEE transactions on cybernetics (IF: 10.50)***, 50(12), 4946-4957, 2019.
25. X. Yu, Z. Zheng, **L. Chen**, Z. Qin, M. He, R. Luo. Leveraging ONT move table values for signal aware variant calling. bioRxiv, 2026-02. (Submitted to ***ECCB 2026***)

Granted patents

1. IMPLEMENTATION METHOD, DEVICE AND EQUIPMENT OF ADAPTIVE PROTEIN PREDICTION FRAMEWORK. (2021)
   - CN patent no. CN202110600871.7. Lei Chen, Min Yang, ..., Qingshan Jiang.
2. A METHOD FOR OBTAINING A RESOURCE RECOMMENDATION MODEL, METHOD, APPARATUS, AND DEVICE. (2022)
   - CN patent no. CN202210506185.8. Lei Chen, Jingtao Ding, ..., Lingling Yi.
3. A KNOWLEDGE DISTILLATION METHOD FOR SEQUENTIAL RECOMMENDATION BASED ON EARTH MOVER'S DISTANCE. (2020)
   - CN patent no. CN202011245696.6. Lei Chen, Min Yang, ..., Qingshan Jiang.
4. A METHOD FOR CONSTRUCTING SEQUENTIAL RECOMMENDATION MODEL AND SEQUENTIAL RECOMMENDATION METHOD. (2020)
   - CN patent no. CN202010844315.X. Lei Chen, Min Yang, ..., Qingshan Jiang.
5. A SEQUENTIAL RECOMMENDATION METHOD AND SYSTEM BASED ON ADAPTIVE NETWORK DEPTH. (2020)
   - CN patent no. CN202010835626.X. Lei Chen, Min Yang, ..., Qingshan Jiang.
6. AN ADAPTIVE OUTPUT SEQUENTIAL RECOMMENDATION METHOD AND SYSTEM. (2020)
   - CN patent no. CN202010830579.X. Lei Chen, Min Yang, ..., Qingshan Jiang.
7. A SEQUENTIAL RECOMMENDATION METHOD AND SYSTEM BASED ON IMPROVED RESNET STRUCTURE. (2020)
   - CN patent no. CN202010816525.8. Lei Chen, Min Yang, ..., Qingshan Jiang.
8. A CROSS-DOMAIN SEQUENTIAL RECOMMENDATION METHOD BASED ON ADAPTIVE FINE-TUNING STRATEGY. (2019)
   - CN patent no. CN201911272877.5. Lei Chen, Min Yang, ..., Chengming Li.

Academic services

- Head teaching assistant for *COMP2501 Introduction to Data Science, HKU*. 2022-2026.
- Reviewer for journals and conferences including *Neurocomputing, Knowledge-based Systems, NeurIPS, ACL, EMNLP, AAAI, COLING, ACM-BCB,* etc.
- Organizer of *NLPCC-2020 Shared Task: Multi-Aspect-based Multi-Sentiment Analysis (MAMS)*.
- Volunteer of *GIW XXXIV ISCB-Asia 2025, AAAI-2021,* etc.